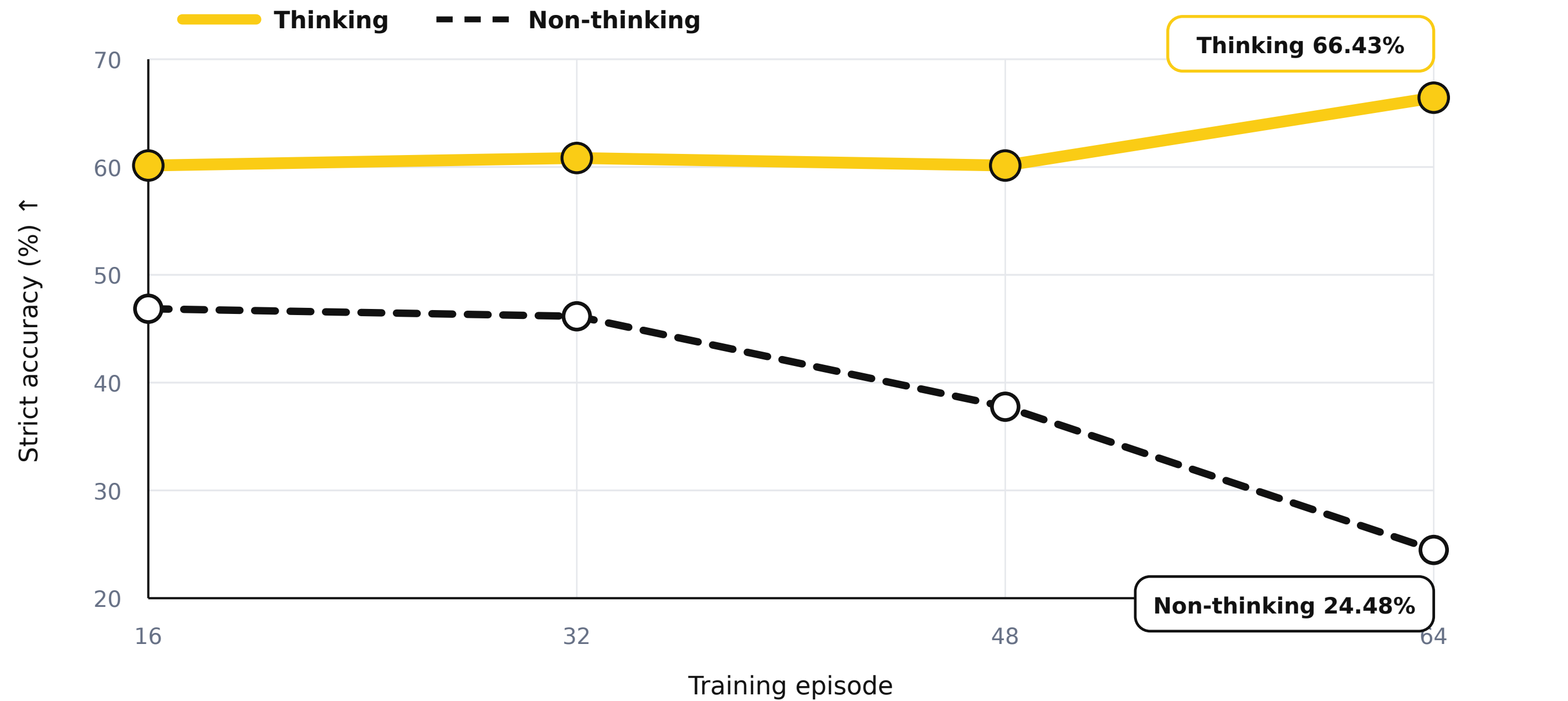


Why Qwen3-8B uses thinking mode

DeepMath-64 training → AMC23/AIME24/AIME25 · 143 problems · strict accuracy

Thinking preserves accuracy on multi-step math

same model · same training method · same seed · same decoding seed



WHY THIS PROTOCOL

- 1 Multi-step demand**

AMC/AIME solutions require dependent algebra, case analysis, and answer verification.
- 2 Reasoning workspace**

Thinking mode gives Qwen3-8B room for intermediate steps before committing to a final answer.
- 3 Observed consequence**

Thinking rises to 66.43%; non-thinking falls to 24.48% after 64 training episodes.